Project Design Phase-II Data Flow Diagram & User Stories

Quotes Recommendation Chatbot using Rasa NLU

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Team ID	SmartBridge-EL-2026
Project Name	Quotes Recommendation Chatbot using Rasa NLU
Maximum Marks	As per evaluation criteria

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Data Flow Diagram (DFD) — Level 0 (Context Diagram)

The DFD below shows how data enters and exits the Quotes Recommendation Chatbot system at the highest level.

Return path: Quote response travels back: Actions → Rasa → Flask → Browser (displayed in chat UI)

DFD — Level 1 (Detailed Process Flow)

Step	Process	Data In	Data Out
1	User sends message via browser	Text string: e.g. 'give me a motivation quote'	HTTP POST request to Flask /send endpoint
2	Flask forwards to Rasa	JSON {sender: user, message: text}	POST to Rasa REST webhook at localhost:5005
3	Rasa NLU classifies intent	Raw text message	Intent label: e.g. motivational_quote, funny_quote, greet
4	Rasa triggers custom action	Intent + entities	Action name: e.g. action_motivational_quote
5	Actions.py fetches quote	Action name	Quote string selected from domain.yml responses
6	Response returned to Flask	JSON {text: quote_string}	Flask receives JSON response from Rasa
7	Flask renders reply	Quote JSON	HTML chat bubble displayed to user in browser

User Stories

User Type	Epic (FR)	USN	User Story / Task	Acceptance Criteria	Priority	Sprint
End User	Greeting	USN-1	As a user, I can say 'hi' or 'hello' to start the chatbot conversation	Bot responds with welcome message listing available categories	High	S-1
End User	Motivational Quote	USN-2	As a user, I can ask for a motivational quote by typing 'motivate me' or 'give me motivation'	Bot returns a relevant motivational quote	High	S-1
End User	Inspirational Quote	USN-3	As a user, I can ask for an inspirational quote in natural language	Bot returns an inspirational quote correctly	High	S-1
End User	Love Quote	USN-4	As a user, I can ask for 'something about love' or a 'love quote'	Bot returns a love-themed quote	High	S-1
End User	Success Quote	USN-5	As a user, I can request a quote about success or achievement	Bot returns a success-themed quote	High	S-1
End User	Funny Quote	USN-6	As a user, I can ask for something funny to lighten my mood	Bot returns a humorous quote	High	S-1
End User	Goodbye	USN-7	As a user, I can say 'bye' to end the conversation	Bot responds with a farewell message	Medium	S-1
End User	Fallback	USN-8	As a user, if the bot doesn't understand my message, I get a helpful prompt	Bot returns fallback message suggesting valid categories	High	S-1
End User	Web Interface	USN-9	As a user, I can interact with the chatbot through a browser at localhost:5000	Chat UI loads, messages send/receive correctly	High	S-1
Developer	Model Training	USN-10	As a developer, I can train the NLU model using 'rasa train' from the project directory	Model trains successfully and is saved in /models	High	S-2
Developer	Testing	USN-11	As a developer, I can run 'rasa test' to validate the model on test stories	All test stories pass with correct intent classification	Medium	S-2

User Type	Epic (FR)	USN	User Story / Task	Acceptance Criteria	Priority	Sprint
Developer	Extensibility	USN-12	As a developer, I can add a new quote category by editing nlu.yml and domain.yml only	New category responds correctly after retraining	Low	S-3